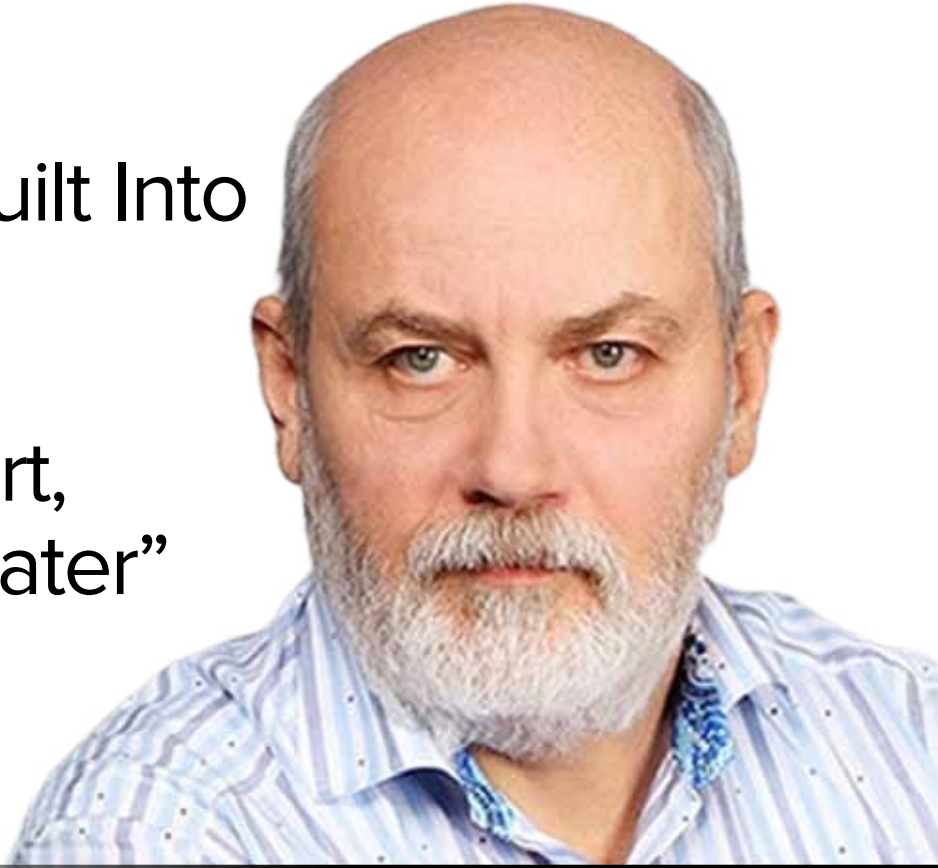


Embedded

“SECURITY Should Be Built Into EMBEDDED SYSTEMS From the Start, Not Added Later”

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How should embedded systems be secured in a connected world? Kaspersky advocates building security in from the ground up. In this exclusive with EFY's Nidhi Agarwal, Andrew Doukhvalov of Kaspersky unpacks their AI-driven approach, open-source strategy, and the thinking behind their cyber-immune architecture.

How is your approach to cybersecurity different from others?

Our core principle is to build strong security for IoT systems during the design and development process—not by layering on security afterwards. We believe security must be integral to how systems are built from the outset. To support this, we offer a new operating system that provides a stable and robust foundation for secure solutions.

How are you using AI and machine learning to enhance cybersecurity?

The key advantage of using artificial intelligence (AI) lies in its ability to make inferences about previously unknown artefacts and events—for example, detecting unusual patterns (called anomalies) in network traffic. It compares these patterns against established norms, both generally and for the specific system. When it detects significant deviations, it alerts system operators so they can investigate and take containment action, if necessary. This is particularly useful in industrial networks, where processes

are highly standardised and tightly controlled.

How does a microkernel help improve security?

Microkernels separate critical components from less critical ones to meet security goals. This principle underpins KasperskyOS, our secure-by-design microkernel operating system. Its architecture enables developers to build trusted systems in which each component operates with minimal, clearly defined privileges—a fundamental